

NONABELIAN COMPOSITION FACTORS OF FINITE GROUPS WHOSE MAXIMAL SUBGROUPS ARE COMPLEMENTED

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ABSTRACT. Let G be a finite group in which every maximal subgroup M has an exact complement: $G = MK$ and $M \cap K = 1$ for some $K \leq G$. We prove that every nonabelian composition factor of G is isomorphic to

$$\mathrm{PSL}_2(7), \quad \mathrm{PSL}_2(11), \quad \text{or} \quad \mathrm{PSL}_5(2),$$

and that all three possibilities occur. This answers Kourovka Problem 18.68. The central reduction passes from a chosen composition factor S to a monolithic quotient with socle S^k , and then constructs a primitive product action of that quotient from an almost-simple coordinate action. A complement to a point stabilizer would be regular, so maximal-factorization and fixed-point arguments can be applied to the coordinate group. The remaining family $\mathrm{PSp}_4(2^f)$ is eliminated by a prime-degree subfield action and a 2-adic valuation argument. GAP tables of marks settle the fifteen remaining finite coordinate groups.

1. INTRODUCTION

Say that a finite group G has property **CMP** if every maximal subgroup $M < G$ has a subgroup $K \leq G$ such that

$$(1) \quad G = MK, \quad M \cap K = 1.$$

The complement is not assumed normal. Kourovka Problem 18.68 asks for the nonabelian composition factors of a nonsoluble group with this property [9, Problem 18.68].

Levchuk and Likharev proved that the nonabelian simple groups having **CMP** are precisely $\mathrm{PSL}_2(7)$, $\mathrm{PSL}_2(11)$, $\mathrm{PSL}_5(2)$ [13, Theorem 1 and Lemmas 4, 6]. Maslova obtained the same composition-factor list under the stronger hypothesis that every maximal subgroup is a Hall subgroup [19, Theorem 1]. Maslova and Revin subsequently described groups under that hypothesis and proved that it implies **CMP** [20, Theorems 1–2]; the implication is not reversible, so those results do not answer the present question. The complete classification of exact factorizations of almost-simple groups [14] addresses the one-coordinate setting, not an ambient monolithic group with socle S^k . We are not aware of a previous solution to Problem 18.68.

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The structural setting has established roots. Aschbacher–Scott organize maximal subgroups through minimal normal subgroups [1]; Kovács studies maximal subgroups of composite finite groups [11] and primitive subgroups of wreath products in product action [12]. We combine these ideas with the complemented-maximal hypothesis: quotient closure places an arbitrary composition factor in a monolithic group, and a selected coordinate action produces a new primitive action of that same quotient.

The obstacle is inheritance. Property **CMP** passes to quotients but not to normal subgroups, hence not automatically to sections. For example, D_8 has **CMP**, while its normal cyclic subgroup C_4 does not. Thus one cannot attach **CMP** directly to a nonabelian composition factor and quote the simple-group theorem.

The reduction is given in [Theorem 2.3](#). Starting from a chosen composition factor S , we pass only to a quotient L whose unique minimal normal subgroup is S^k . The group induced on one simple direct factor is almost simple, $S \leq X \leq \text{Aut}(S)$. A suitable primitive action $X \curvearrowright \Delta$ then yields a faithful primitive product action $L \curvearrowright \Delta^k$. If L has **CMP**, its point stabilizer has a regular complement. Corollary 3(iv) of Liebeck–Praeger–Saxl then reduces the product-action case to a core-free transitive subgroup of the coordinate group [17].

Theorem 1.1. *Let G be a finite group with **CMP**. Every nonabelian composition factor of G is isomorphic to one of*

$$\text{PSL}_2(7), \quad \text{PSL}_2(11), \quad \text{PSL}_5(2).$$

Each of the three possibilities occurs.

The proof uses the classification of finite simple groups and the published classifications of maximal factorizations of almost-simple groups. The ordinary group theory—quotient closure, the monolithic reduction, the product action, and the fixed-point and valuation arguments—is developed in [Sections 2](#) and [3](#). Three representative applications appear in [Section 4](#); the remaining family arguments are given in [Section 5](#) and [Appendix A](#). The finite calculations are recorded in [Appendix B](#).

2. REDUCTION TO A MONOLITHIC QUOTIENT

Let S be a nonabelian composition factor of a group with **CMP**. The first step is to retain the hypothesis by passing to a quotient L with a unique self-centralizing minimal normal subgroup S^k . This avoids assigning **CMP** to the section in which S originally occurs.

The normalizer of one simple direct factor of $\text{Soc}(L)$ induces an almost-simple group X . Rather than seeking a quotient map from L to X , we use the standard wreath embedding to construct a product action of L from a coset action of X . The two steps are carried out below.

2.1. Quotient closure.

Lemma 2.1 (quotient closure). *If G has CMP and $N \trianglelefteq G$, then G/N has CMP.*

Proof. Let $\overline{M}$ be maximal in G/N , and let M be its full preimage. Choose $K \leq G$ satisfying (1). Then $G/N = \overline{M}(KN/N)$. If $kN \in \overline{M}$ with $k \in K$, then $k \in M$, and hence $k = 1$. Thus KN/N is an exact complement to $\overline{M}$. $\square$

Proposition 2.2 (monolithic reduction). *Let S be a nonabelian composition factor of a finite CMP-group G . There is a quotient L of G such that*

- (1) L has CMP;
- (2) $\text{Soc}(L) = N \cong S^k$ for some $k \geq 1$, and N is the unique minimal normal subgroup of L ;
- (3) $C_L(N) = 1$.

Proof. Choose a chief factor $A/B \cong S^k$ of G , and define

$$C/B = C_{G/B}(A/B), \quad L = G/C, \quad N = AC/C.$$

Since A/B is centerless, $A \cap C = B$, and hence $N \cong A/B$. Chiefness makes N a minimal normal subgroup of L . If gC centralizes AC/C , then $[g, A] \leq A \cap C = B$, so $g \in C$ by the definition of $C/B = C_{G/B}(A/B)$. Thus the conjugation action of L on N is faithful and $C_L(N) = 1$. If R were another minimal normal subgroup, then $[R, N] \leq R \cap N = 1$, forcing $R \leq C_L(N) = 1$. Thus N is unique. Finally, L has CMP by Lemma 2.1. $\square$

The example $C_4 \trianglelefteq D_8$ shows why the preceding proposition, rather than section inheritance, is the correct starting point. The next construction turns its entire direct-power socle into a coordinate problem.

2.2. The product action. Write

$$N = S_1 \times \cdots \times S_k \cong S^k.$$

The action of L on the factors is transitive, since the product of the factors in any orbit is normal in L . Set

$$(2) \quad X = N_L(S_1)/C_L(S_1),$$

identifying S_1 with its inner automorphism group. Then $S \leq X \leq \text{Aut}(S)$. The standard wreath embedding gives

$$(3) \quad L \hookrightarrow X \wr P \leq X \wr \text{Sym}(k),$$

where P is transitive, N is the base socle S^k , and the group induced on a coordinate is X . An element in the kernel fixes every factor and induces the identity automorphism on each S_i , so it centralizes N . Thus the kernel lies in $C_L(N) = 1$, and the embedding is faithful.

Let $H < X$ be core-free and maximal. Put $\Delta = X/H$ and $V = H \cap S$, and assume $V \neq 1$. Since S is normal and is not contained in H , we have

$X = SH$; in particular, S is transitive on Δ . Via (3), let L act on Δ^k in product action.

Theorem 2.3 (product-action reduction). *The action of L on Δ^k is faithful and primitive. If $\omega = (H, \dots, H)$ and $M = L_\omega$, then $L = MS^k$ and $M \cap S^k = V^k$.*

Here we apply standard wreath-product and Scott-lemma ingredients. The proof tests an intermediate overgroup through its intersection with S^k , then uses Scott's lemma to show that the only possible strips have length one.

Proof. Let $\omega = (H, \dots, H)$, $M = L_\omega$, and $N = S^k$. The group N is transitive, so $L = MN$ and $M \cap N = V^k$.

Suppose $M \leq J \leq L$, and put $D = J \cap N$. Then $J = MD$, $D \trianglelefteq M$, and $V^k \leq D$. The image of M on the coordinates is transitive: multiply an element of L by a suitable base element to make it fix ω without changing its top permutation. Moreover, the subgroup of M stabilizing coordinate i induces H there. For a prescribed $h \in H$, lift h through $N_L(S_i) \rightarrow X$. Its i -coordinate already fixes the coset H , and the transitivity of each other factor permits multiplication by suitable elements of S_j , one for each $j \neq i$, that return all remaining coordinates to H . The corrected lift lies in M and still induces h on S_i .

Let D_i be the projection of D to $S_i \cong S$. It contains V and is normalized by H . Maximality of H implies either $D_i = V$ or $HD_i = X$. In the second case

$$D_i = (H \cap S)D_i = (HD_i) \cap S = X \cap S = S.$$

Coordinate transitivity gives the same alternative for every i . If all projections are V , then $D = V^k$ and $J = M$. Otherwise D is subdirect in S^k . By Scott's lemma [21, Lemma, p. 328] it is a direct product of full diagonal strips. Since $V^k \leq D$ and $V \neq 1$, every strip has length one, so $D = N$ and $J = L$. Thus M is maximal. Faithfulness follows from core-freeness of H and (3). $\square$

The theorem removes the unbounded exponent k . It constructs a new action of the same quotient L ; it does not assume that the original primitive action supplied by CMP already has product-action type. The next section explains how the existence of a complement in this action constrains the one-coordinate group X .

3. OBSTRUCTIONS TO REGULAR COMPLEMENTS

Let $H < X$ be as in Theorem 2.3, put $\Delta = X/H$, and write $V = H \cap S \neq 1$. A complement to the lifted point stabilizer is a regular subgroup. We shall exclude such a subgroup using factorizations of X , fixed points of elements of prime order, or a valuation comparison between the degree and $|X : S|$.

Throughout, X/H denotes the set $\{Hx : x \in X\}$ of right cosets, on which X acts by right multiplication. Dotted Atlas notation such as $M_{12.2}$

denotes an extension without asserting that it splits; a colon denotes a split extension, and subscripts distinguish Atlas isomorphism types.

Lemma 3.1 (regular-action reformulation). *A subgroup $K \leq G$ satisfies (1) if and only if it is regular on the right cosets of M . If M is maximal and core-free, this is a faithful primitive action.*

Proof. Transitivity is equivalent to $G = MK$, and the stabilizer in K of the coset M is $M \cap K$. $\square$

3.1. Factorizations of the coordinate group. Call (X, Δ) *factor-free* if it has no core-free transitive subgroup; equivalently, there is no core-free $C < X$ with $X = HC$.

Proposition 3.2 (factorization obstruction). *If $(X, X/H)$ is factor-free, then L does not have CMP.*

Proof. For $k = 1$, we have $L = X$. If a regular subgroup R existed, then R would be transitive and $R \cap H = 1$. Its core in X , if nontrivial, would contain the socle S , as X is almost simple. But then $S \leq R$, contrary to $1 \neq H \cap S \leq H$ and $R \cap H = 1$. Thus R would be core-free, contradicting factor-freeness. Let $k \geq 2$. The action in [Theorem 2.3](#) is a primitive product action with almost-simple component $(X, X/H)$: indeed, its socle stabilizer is V^k , which is nontrivial and is not subdirect in S^k . Thus the action is neither affine nor of diagonal or twisted-wreath type; in the notation of [\[17, Corollary 3\(iv\)\]](#), its almost-simple component is precisely X acting on X/H . That corollary states that a primitive group of this product-action type with a regular subgroup has a core-free transitive subgroup in its component action. Factor-freeness therefore excludes a regular subgroup of L . If L had CMP, its maximal point stabilizer would have a complement, regular by [Lemma 3.1](#), a contradiction. $\square$

We next reduce factor-freeness to the published lists of maximal factorizations.

Lemma 3.3 (maximal-factorization reduction). *Let $1 < V < S$, and assume that the S -conjugacy class of V is invariant under X . Choose a maximal subgroup $H < X$ containing $N_X(V)$. Then H is core-free and $X = HS$. If X/H is not factor-free, there are $S \leq Y \leq X$ and a nontrivial maximal factorization $Y = AB$ such that, after interchanging the factors,*

$$V \leq A \cap S.$$

Proof. Class invariance gives $N_X(V)S = X$, so no proper overgroup of $N_X(V)$ contains S . Hence H is core-free and $HS = X$.

Suppose $C < X$ is core-free and $X = HC$. Put $Y = CS$. Then $Y = (H \cap Y)C$, and both factors supplement S in Y . Consequently all their proper maximal overgroups are core-free. Enlarging them gives a maximal factorization $Y = AB$, with $V \leq H \cap S \leq A \cap S$. $\square$

3.2. Elusive permutation actions.

Definition 3.4. An action of Z on Λ is *p-elusive* if $p \mid |\Lambda|$ and every element of order p in Z fixes a point.

Lemma 3.5 (wreath lifting). *If X is p-elusive on Δ , then $X \wr \text{Sym}(k)$ is p-elusive on Δ^k . In particular, no subgroup of this wreath product is regular on Δ^k .*

Proof. Let $w = (x_1, \dots, x_k)\sigma$ have order p . On a fixed coordinate of σ , the corresponding x_i has order 1 or p , and hence fixes a point. On a p -cycle of σ , the cycle product of the x_i is 1, since $w^p = 1$; the fixed-tuple equations can therefore be solved successively around the cycle. Thus w fixes a tuple.

If a subgroup were regular, Cauchy's theorem would give it a nonidentity element of order p , because its order is $|\Delta|^k$. That element would fix a point, a contradiction. $\square$

3.3. A valuation criterion.

Lemma 3.6 (socle-valuation criterion). *Suppose S is p-elusive on $\Delta = X/H$. Let*

$$a = v_p(|\Delta|), \quad o = v_p(|X : S|).$$

If $a > o$, then every subgroup $L \leq X \wr \text{Sym}(k)$ containing S^k contains no regular subgroup on Δ^k .

Proof. Suppose $R \leq L$ is regular and put $Q = R \cap S^k$. Since $R/Q \hookrightarrow L/S^k$ and $L/S^k \leq (X/S)^k \rtimes \text{Sym}(k)$,

$$v_p(|R : Q|) \leq ko + v_p(k!).$$

As $v_p(|R|) = ka$,

$$v_p(|Q|) \geq k(a - o) - v_p(k!) > 0,$$

because $a - o \geq 1$ and $v_p(k!) < k$. An element of order p in Q fixes a tuple coordinatewise, contrary to regularity. $\square$

4. APPLICATIONS TO THREE FAMILIES

The following examples exhibit the three arguments used later in the classification: fixed points of prime-order elements, maximal factorizations, and a valuation comparison.

4.1. The two-subset action of A_5 .

Proposition 4.1. *For $A_5 \leq X \leq S_5$, the action of X on the ten 2-subsets of $\{1, \dots, 5\}$ is 2-elusive. Its point stabilizer is core-free and maximal, and has nontrivial intersection with A_5 .*

Proof. Write H for the stabilizer of a 2-subset. It is maximal in S_5 , and $H \cap A_5$ is maximal in A_5 ; the actions are therefore primitive and faithful. Every involution in S_5 , and hence in either coordinate group, contains a 2-cycle and fixes its support as a 2-subset. Since $2 \mid 10$, the action is 2-elusive. The subgroup $H \cap A_5$ contains a double transposition and is nontrivial. $\square$

By Lemma 3.5, this single fixed-point observation excludes a regular complement on every Cartesian power.

4.2. Six-subset actions of alternating groups.

Proposition 4.2. *Let $n \geq 13$, let $A_n \leq X \leq S_n$, and let H stabilize a 6-subset. Then H is core-free and maximal, $H \cap A_n \neq 1$, and X/H is factor-free.*

Proof. The stabilizer is maximal because $6 < n/2$. Suppose that $C < X$ is core-free and $X = HC$. Apply Liebeck–Praeger–Saxl [16, Theorem D and Remark 2] with $L = A_n \triangleleft X \leq \text{Aut}(A_n)$, $A = H$, and $B = C$. Both factors satisfy the theorem’s hypothesis of not containing L . Since $n \geq 13$, none of the exceptional degrees 6, 8, 10 occurs, and the theorem gives

$$A_{n-j} \leq H \leq (S_{n-j} \times S_j) \cap X \quad (1 \leq j \leq 5),$$

while C is j -homogeneous in the natural action. Maximality of H forces $H = (S_{n-j} \times S_j) \cap X$, which is impossible because its two natural orbit sizes are 6 and $n - 6$, whereas $j \leq 5$. Thus no such C exists. Core-freeness follows because every nontrivial normal subgroup of X contains A_n , whereas H does not; moreover $H \cap A_n$ contains $A_6 \times A_{n-6}$. $\square$

4.3. Subfield actions of $\text{PSp}_4(2^f)$.

Proposition 4.3. *Let $S = \text{PSp}_4(2^f)$, $f \geq 2$, and $S \leq X \leq \text{Aut}(S)$. There is a core-free maximal subgroup $H < X$ with $H \cap S \neq 1$ such that S is 2-elusive on X/H and*

$$v_2(|X : H|) > v_2(|X : S|).$$

Consequently the lifted product action has no regular subgroup for any socle multiplicity $k \geq 1$.

Proof. Let $q = 2^f$. Choose a prime $r \mid f$, put $d = f/r$, $q_0 = 2^d$, and let

$$V = \text{Sp}_4(q_0) < S$$

be the standard subfield subgroup. It is maximal in S , since the field-extension degree r is prime [2, Table 8.14]; see also [3, Proposition 4.2 and Table 3].

We check explicitly that this one choice works for every field-only and graph-outer coordinate group $S \leq X \leq \text{Aut}(S)$. Realize S as the fixed points of the standard Frobenius φ^f in the adjoint algebraic group of type B_2 over $\overline{\mathbb{F}}_2$. The exceptional graph-field endomorphism ρ satisfies $\rho^2 = \varphi$, as in the setup preceding Lemma 2.1 and Theorem 4(a) of [7], while $\text{Aut}(S) = \langle S, \rho \rangle$ by [7, Lemma 2.1]. Since ρ commutes with φ , it normalizes the fixed-point subgroup $V = \text{Sp}_4(2^d)$. Thus the S -class of V is invariant under all of $\text{Aut}(S)$.

Put $H = N_X(V)$. Class invariance gives $X = SH$, and $H \cap S = N_S(V) = V$. Moreover, H is maximal in X . Indeed, if $H < K < X$, then maximality of V in S gives $K \cap S = V$ or S . In the first case, $V = K \cap S \trianglelefteq K$, so

$K \leq N_X(V) = H$; in the second, $S \leq K$ and $X = SH \leq K$. Both are contradictions. Finally, H is core-free: every nontrivial normal subgroup of the almost-simple group X contains S , whereas $H \cap S = V < S$.

Every involution class of S meets V . Here are the details needed at the prime-field endpoint $q_0 = 2$. By Lemmas 2.1–2.2 and Corollary 2.3 of [18], the three involution classes of S meet

$$O_4^+(q) \cong (\mathrm{SL}_2(q) \times \mathrm{SL}_2(q)) \rtimes \langle \tau \rangle,$$

where τ interchanges the factors, and conjugacy inside this orthogonal group agrees with conjugacy in S . If u represents the unique nonidentity involution class in $\mathrm{SL}_2(q)$, the three orthogonal classes have representatives $(u, 1), (u, u), \tau$. We may take $u \in \mathrm{SL}_2(2)$, so all three representatives lie in $O_4^+(2) \leq \mathrm{Sp}_4(2) \leq V$. Hence S is 2-elusive on X/H .

Since $X = SH$, the degree and its 2-part are

$$(4) \quad |X : H| = |S : V| = \frac{q^4(q^2 - 1)(q^4 - 1)}{q_0^4(q_0^2 - 1)(q_0^4 - 1)}, \quad a = v_2(|X : H|) = 4(f - d).$$

The same outer-automorphism description gives $|X : S| \mid 2f$, so

$$o = v_2(|X : S|) \leq 1 + v_2(f) < 2f \leq 4(f - d) = a.$$

Here $v_2(f) \leq f - 1$, and $f - d = f(1 - 1/r) \geq f/2$. Thus [Lemma 3.6](#) applies uniformly. In particular, when $q = 4$ and $X = \mathrm{Aut}(S)$, the subgroup has $|H| = 2880$, $|H \cap S| = 720$, and degree 1360; these values are also reproduced in [Appendix B](#). $\square$

Thus a fixed-point statement for the socle is enough whenever the outer 2-part is smaller than the 2-part of the degree.

5. CLASSIFICATION OF THE COORDINATE GROUPS

Proposition 5.1. *Let S be a nonabelian finite simple group not isomorphic to $\mathrm{PSL}_2(7)$, $\mathrm{PSL}_2(11)$, or $\mathrm{PSL}_5(2)$, and let $S \leq X \leq \mathrm{Aut}(S)$. There is a core-free maximal subgroup $H < X$, with $H \cap S \neq 1$, for which one of the following holds:*

- (1) X/H is factor-free;
- (2) X is p -elusive on X/H for some prime p ;
- (3) $S = \mathrm{PSp}_4(2^f)$, $f \geq 2$, and the hypotheses of [Lemma 3.6](#) hold with $p = 2$.

Proof. The common argument is to choose an automorphism-invariant subgroup $1 < V < S$, place $N_X(V)$ in a core-free maximal subgroup H , and apply [Lemma 3.3](#). The following table identifies every family, the subgroup choices and exceptions, and the location of the details.

CFSG family	subgroup choices and exceptions	details
alternating	small degrees by explicit elusive actions or five finite computations; $n \geq 13$ by 6-subsets	Proposition 4.1 and Proposition 4.2 ; Appendix A.1
classical	invariant parabolics, torus normalizers, or geometric subgroups; $L_3(4)$ is finite and even PSp_4 uses valuation	Proposition 4.3 ; Appendix A.2 and Appendix B
exceptional Lie type	defining-characteristic Sylow or graph-stable parabolic intersections	Appendix A.3
sporadic	Atlas maximal subgroups checked against Giudici's exhaustive factorization tables; M_{24} is 2-elusive	Appendix A.4

The alternating, classical, exceptional, and sporadic groups exhaust the nonabelian finite simple groups. The appendix includes every outer coordinate group and low-parameter isomorphism. Together with the three arguments above, this establishes the proposition. $\square$

The classification is used only through this proposition; the case-by-case details are separated into [Appendix A](#).

6. PROOF OF THE MAIN THEOREM

Proof of [Theorem 1.1](#). Let S be a nonabelian composition factor of G . Apply [Proposition 2.2](#) to obtain a CMP-quotient L with unique self-centralizing socle S^k , and define its coordinate group X by [\(2\)](#).

Suppose S is not one of the three groups in the conclusion. Choose the maximal subgroup $H < X$ supplied by [Proposition 5.1](#). By [Theorem 2.3](#), the same group L has a faithful primitive action on $(X/H)^k$. If the action X/H is factor-free, [Proposition 3.2](#) contradicts $\mathrm{CMP}(L)$. If it is p -elusive, [Lemma 3.5](#) shows that the primitive action has no regular subgroup, again contradicting $\mathrm{CMP}(L)$. In the final symplectic case, the same contradiction follows from [Lemma 3.6](#). Therefore S is one of the three displayed groups.

Conversely, Levchuk–Likharev prove that each of the three simple groups has CMP [[13](#), Lemmas 4 and 6]; taking the group itself shows that each occurs as a composition factor. $\square$

7. CONCLUSION

The proof teaches how to constrain a composition factor when a maximal-subgroup property is quotient-closed but not section-closed. The monolithic quotient keeps the ambient hypothesis intact, while [Theorem 2.3](#) converts its direct-power socle into a family of coordinate actions. This use of established

monolithic and product-action machinery is CMP-specific: it replaces an invalid inheritance assertion by a primitive action whose point stabilizer is still a maximal subgroup of the original quotient.

The essential hypotheses are visible in the proof. Quotient closure is needed to reach L ; the unique self-centralizing socle gives a faithful wreath embedding; a nontrivial coordinate intersection $H \cap S$ forces primitivity; and the maximal-subgroup property must turn the lifted point stabilizer into a regular subgroup. Other quotient-closed properties that force such a regular or suitably transitive partner may admit a similar reduction.

The final list depends on CFSG and the classifications of maximal factorizations only through [Proposition 5.1](#). The product-action reduction explains why almost-simple factorizations control arbitrary powers S^k , while the symplectic calculation shows how a weaker fixed-point property can be amplified by a valuation inequality. A classification-free proof, or an application of the same reduction to another maximal-subgroup property, remains a natural direction.

APPENDIX A. CLASSIFICATION DETAILS

This appendix supplies the family arguments used in [Proposition 5.1](#). Each subsection gives the common subgroup choice, treats the exceptional coordinate groups, and cites the relevant classification theorem.

A.1. Alternating groups. For $n \neq 6$, the coordinate group is A_n or S_n . The following actions are p -elusive:

Socle	action and degree	prime
A_5	2-subsets, 10	2
$A_6, X = A_6$	natural, 6	2
$A_6, X = S_6$	unordered 3 + 3 partitions, 10	2
A_8	2-subsets, 28	2
A_9	2-subsets, 36	2
A_{10}	4-subsets, 210	2
A_{11}	3-subsets, 165	3
A_{12}	2-subsets, 66	2

These assertions follow from the cycle supports. An involution fixes the support of one 2-cycle. In degree 10, every involution also has an invariant 4-set: use two 2-cycles, or one 2-cycle and two fixed points. An element of order 3 in degree 11 fixes the support of one 3-cycle. For the S_6 partition action, the cycle types $2, 2^2, 2^3$ respectively preserve a 3-set or interchange it with its complement.

The three outer extensions of A_6 not equal to S_6 , and the two groups A_7, S_7 , have factor-free maximal subgroups verified in [Appendix B](#). Their

orders and indices are

X	$A_{6.2_2}$	$A_{6.2_3}$	$A_{6.2^2}$	A_7	S_7
$ H $	16	20	32	72	144
$ X : H $	45	36	45	35	35.

For $n \geq 13$, the factor-free action on 6-subsets is proved in [Proposition 4.2](#).

A.2. Classical groups. We use Theorem 2.15 and Appendix Tables A.1–A.7 of Xia–Li [15], a restatement of the LPS classification of nontrivial maximal factorizations with classical socle, together with the two corrected $\text{P}\Omega_8^+(4)$ and $\text{P}\Omega_8^+(16)$ rows of Gill–Giudici–Spiga [5, Section 2]. In the table below, V is invariant up to S -conjugacy under X . Applying [Lemma 3.3](#), inspection of the indicated factorization table shows that no factor intersection contains V .

We follow the table notation of Xia–Li: P_i is the standard maximal parabolic associated with node i , $P_{i,j} = P_i \cap P_j$ is the intersection of the indicated standard parabolics containing a common Borel subgroup, and N_i denotes the relevant nonsingular i -space stabilizer. The phrase “full $2 + 2$ decomposition stabilizer” includes the element interchanging the two summands.

Socle and conditions	subgroup V	factorization-table check
$\text{PSL}_2(q)$, excluding the allowed and alternating cases	split-torus normalizer	not in the nonsplit-torus factor of A.1:1 or in P_1 ; A.1:5–8 are checked below
$\text{PSL}_3(q)$, $q \neq 4$, excluding $q = 2$	split-torus normalizer	not in either parabolic or the extension-field factor in A.1:1
$\text{PSL}_3(4)$, every coordinate group X	a maximal subgroup of X	computational verification in Appendix B
$\text{PSL}_n(q)$, even $n \geq 4$	$P_{n/2}$	absent from A.1:1–4 (and A.1:5–10 have other dimensions)
$\text{PSL}_n(q)$, odd $n \geq 5$, no graph	P_2	absent from A.1:1–4,10 except $(n, q) = (5, 2)$
$\text{PSL}_n(q)$, odd $n \geq 5$, graph present	$P_{2,n-2}$	only maximal parabolic overgroups P_2, P_{n-2} ; A.1:1–4,10 are excluded except $(5, 2)$
$\text{PSp}_{2m}(q)$, $m \geq 3$	P_2	absent from A.2:1–16 (A.2:13 has P_2 only when $m = 2$)

Socle and conditions	subgroup V	factorization-table check
$\mathrm{PSp}_4(q)$, q odd	nonsingular $2 + 2$ decomposition stabilizer	absent from the $m = 2$, odd- q rows A.2:1,13
$\mathrm{PSU}_3(q)$	N_1	absent from the $n = 3$ rows A.3:5–7
$\mathrm{PSU}_4(q)$, $q \geq 3$	full nonsingular $2 + 2$ decomposition stabilizer	absent from A.3:1–4,9; in particular A.3:4 covers $\mathrm{PSU}_4(4)$
$\mathrm{PSU}_n(q)$, $n \geq 5$	N_2	absent from A.3:1–4,10–12
$\Omega_{2m+1}(q)$, $m \geq 3$, q odd	P_2	absent from A.4:1–8
$\mathrm{P}\Omega_{2m}^-(q)$, $m \geq 4$	P_2	absent from A.5:1–5
$\mathrm{P}\Omega_{2m}^+(q)$, $m \geq 5$	P_2	absent from A.6:1–13
$\mathrm{P}\Omega_8^+(q)$, triality allowed	central-node P_2	fixed by triality and absent from A.7:1–15

The two corrected orthogonal factorizations have socle intersections N_2^- and a triality image of $\Omega_8^-(q^{1/2})$, for $q = 4, 16$. Neither contains the central-node maximal parabolic P_2 , so they do not alter the last row of the table.

Here a standard maximal parabolic can only be contained in a proper factor intersection if it is that intersection. A subgroup containing $P_{2,n-2}$ contains a Borel and is therefore parabolic; its only proper overgroups are P_2 and P_{n-2} . These observations justify the containment, rather than mere equality, assertions in the table. The displayed N_i and full $2 + 2$ decomposition stabilizers are standard maximal geometric subgroups for the stated parameters [10, Chapters 3–4]; the exceptional $\mathrm{PSU}_4(2)$ parameter is covered by the low-rank isomorphism below.

For the first row, the split-torus normalizer has order $2(q-1)/(2, q-1)$ and interchanges the two fixed points of its torus, so it is not contained in P_1 . Nor can it lie in the nonsplit-torus normalizer: its order would force $q-1 \mid q+1$, apart from nonsimple parameters. The other A.1 factors occur only for

$$q = 7, 11, 16, 19, 23, 29, 59.$$

The values 7, 11 are allowed. At $q = 19, 29, 59, 23$, divisibility by the torus-normalizer order excludes A_5 or S_4 . At $q = 16$, a subgroup of order 30 lies neither in D_{34} nor in A_5 , the latter because it would have index 2. This closes the rank-one family.

For $S = \mathrm{PSL}_3(q)$, let $d = (3, q-1)$ and let V normalize a maximally split torus. Its class is invariant under diagonal, field, and graph automorphisms. The monomial action shows that V fixes neither a point nor a hyperplane.

Its order is $|V| = 6(q-1)^2/d$. The extension-field factor in row 1 of Table A.1 has intersection with S of order $3(q^2 + q + 1)/d$. For $q = 3$, divisibility excludes containment; for $q \geq 5$, the former order is larger. The only remaining table row is the exceptional $q = 4$ row, and all ten almost-simple groups with that socle are handled by the finite computation. The parameter $q = 2$ gives the allowed group $\text{PSL}_3(2) \cong \text{PSL}_2(7)$.

The even four-dimensional symplectic family is the sole infinite row not closed by the maximal-factorization argument itself. It is eliminated uniformly by the prime-degree subfield action in [Proposition 4.3](#). The relevant low-rank isomorphisms are

$$\begin{aligned} \text{PSL}_2(4) &\cong \text{PSL}_2(5) \cong A_5, & \text{PSL}_2(9) &\cong A_6, \\ \text{PSL}_3(2) &\cong \text{PSL}_2(7), & \text{PSL}_4(2) &\cong A_8, \\ \text{PSU}_4(2) &\cong \text{PSp}_4(3). \end{aligned}$$

The identifications $B_2 = C_2$, $D_3 = A_3$, and ${}^2D_3 = {}^2A_3$ place every remaining overlap in a row already considered.

A.3. Exceptional groups of Lie type. Hering–Liebeck–Saxl classify all such factorizations [[8](#), Theorems 1–2]. The only socles admitting a core-free factorization are

$$G_2(4), \quad G_2(3^f), \quad F_4(2^f),$$

together with one extra outer factorization for $G_2(4).2$. None of the listed factors is parabolic. For every exceptional socle outside these three families, take V to be a Sylow subgroup in the defining characteristic. Its S -conjugacy class is invariant under every automorphism, and the Hering–Liebeck–Saxl theorem and [Lemma 3.3](#) make the resulting action factor-free.

It remains to treat the three factorable families. If no graph automorphism is present, choose a maximal parabolic. In the graph-fused cases choose the graph-stable parabolic intersection $P_1 \cap P_2$ in $G_2(3^f)$, or $P_1 \cap P_4$ in $F_4(2^f)$; each contains a common Borel. For the graph-fused $G_2(3^f)$ branch, the chosen subgroup has p -part q^6 , whereas the displayed Hering–Liebeck–Saxl factors have p -part at most q^3 ; the group $G_2(4)$ is covered separately by its listed factorization. In the $F_4(2^f)$ branch, the corresponding bounds are at least q^{24} and at most q^{16} . Thus no factor contains the chosen subgroup. The small isomorphisms $G_2(2)' \cong \text{PSU}_3(3)$ and ${}^2G_2(3)' \cong \text{PSL}_2(8)$ have already been covered.

A.4. Sporadic groups. Giudici classifies all factorizations with sporadic socle [[6](#), Theorems 1.1–1.3 and Tables 1–4]. Theorem 1.1 and Tables 1–2 treat simple sporadic groups; Theorem 1.2 and Table 3 give the new outer cases; and Theorem 1.3 and Table 4 list the exact factorizations. The factor-free argument uses the complete factorization lists in Tables 1–3, not the exact-only list in Table 4. Comparing those loci with the Atlas maximal-subgroup lists [[4](#), [22](#)] gives the following factor-free maximal subgroups. A

row for an outer extension also excludes the factorizations lifted from the simple socle by Giudici’s Lemma 2.1(5).

X	H	X	H
M_{11}	S_5	M_{12}	$2^3:S_4$
$M_{12}.2$	$3^{1+2}:D_8$	$M_{22}.2$	$A_6.2^2$
M_{23}	A_8	J_2	$3.A_6.2_2$
$J_2.2$	$3.A_6.2^2$	HS	$L_3(4).2_1$
$HS.2$	$L_3(4).2^2$	He	$2^2.L_3(4).S_3$
$He.2$	$2^2.L_3(4).D_{12}$	Ru	$2^6:U_3(3):2$
Suz	$J_2.2$	$Suz.2$	$J_2.2 \times 2$
Fi_{22}	$O_7(3)$	$Fi_{22}.2$	$G_2(3).2$
Co_1	$2^{11}:M_{24}$		

The simple group M_{22} , and every sporadic almost-simple group not needed in this display other than M_{24} , has no nontrivial core-free factorization. For such a group, apply [Lemma 3.3](#) to a Sylow subgroup of the socle; its conjugacy class is automorphism-invariant and its maximal overgroup has non-trivial socle intersection. The rows available in TomLib are also checked in [Appendix B](#).

The remaining case is M_{24} . Let it act on the 2-subsets of its natural 24-point set. The point stabilizer $M_{22}.2$ is maximal, the degree is 276, and every involution fixes the support of one of its 2-cycles. Thus the action is 2-elusive and [Lemma 3.5](#) applies.

For the eight displayed rows not present in the TomLib calculation, maximality follows from the “Maximal subgroups” sections of the official online ATLAS pages: [He](#), [He.2](#), [Ru](#), [Suz](#), [Suz.2](#), [Fi₂₂](#), [Fi₂₂.2](#), and [Co₁](#).

The four subsections cover every CFSG family, every almost-simple coordinate group, and every low-rank overlap, proving [Proposition 5.1](#).

APPENDIX B. COMPUTATIONAL VERIFICATION OF THE FINITE CASES

The classification argument leaves five small alternating coordinate groups and the ten almost-simple coordinate groups with socle $\text{PSL}_3(4)$. For each group X , the calculation chooses the maximal subgroup H listed below and verifies that there is no core-free subgroup $C < X$ satisfying $X = HC$.

For a complete table of marks T , let h, c, i denote the conjugacy classes of subgroups H, C, I . The entry `IntersectionsTom(T,h,c)[i]` is positive exactly when suitable conjugates of H and C have intersection in class i . Such conjugates factor X precisely when

$$|H||C| = |X||I|.$$

The calculation examines every subgroup class c not containing the socle and every possible intersection class i . It also verifies that $H \cap S \neq 1$, where $S = \text{Soc}(X)$.

For the alternating groups, the subgroups and degrees are

X	$A_{6.2_2}$	$A_{6.2_3}$	$A_{6.2^2}$	A_7	S_7
$ H $	16	20	32	72	144
$ X : H $	45	36	45	35	35.

Since $\text{Out}(\text{PSL}_3(4)) \cong D_{12}$, the following ten TomLib names exhaust, up to isomorphism, the coordinate groups between the socle and its full automorphism group:

X	$ H $	$ X : H $	X	$ H $	$ X : H $
$L_3(4)$	960	21	$L_3(4).2_1$	384	105
$L_3(4).2_2$	1920	21	$L_3(4).2_3$	720	56
$L_3(4).3$	216	280	$L_3(4).2^2$	768	105
$L_3(4).3.2_2$	432	280	$L_3(4).3.2_3$	1152	105
$L_3(4).6$	1152	105	$L_3(4).D_{12}$	2304	105.

In all fifteen cases the number of core-free factor classes is zero.

The computation is in the accompanying repository <https://github.com/RichieSater/kourovka-18-68>. It uses GAP 4.15.1 and TomLib 1.2.11 and is reproduced by

```
gap --quitionbreak -q gap/generate-factor-free-scan.g
python3 tests/check-factor-free-scan.py
```

The resulting data are in `data/tomlib-factor-free.tsv`. The same calculation was also run for

$$M_{11}, M_{12}, M_{12.2}, M_{22.2}, M_{23}, J_2, J_{2.2}, HS, HS.2,$$

providing independent finite checks of the corresponding sporadic rows in [Appendix A.4](#). Finally, a GAP 4.15.1/AtlasRep 2.1.11 calculation checks the $q = 4$ instance in [Proposition 4.3](#), including $|H| = 2880$, $|H \cap S| = 720$, and $|X : H| = 1360$; the general symplectic argument is the proof in [Proposition 4.3](#).

GENERATIVE-AI DISCLOSURE

Generative-AI tools were used for literature search, proof exploration, code generation, drafting, and critical checking; the author checked and takes responsibility for the manuscript's arguments, citations, and computations.

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